Evaluating and Improving Large Language Model Accuracy for Occupational Cancer Information Using the ILO Diagnostic and Exposure Criteria

Jeremy Qiao[0009-0000-0375-3194]

University of New Brunswick, Saint John, NB, Canada
g9rbj@unb.ca

Abstract. This project tests and improves how well Large Language Models (LLMs) answer questions about occupational cancer using the ILO Diagnostic and Exposure Criteria for Occupational Diseases (2022). A major problem with LLMs is hallucination — they sometimes produce answers that sound correct but are factually wrong. This paper builds an Ontology-Guided Retrieval-Augmented Generation (Ontology RAG) system to reduce this problem. The system uses a structured knowledge base (ontology) built from Section 3 of the ILO document (pages 524–606), covering 20 occupational carcinogens. The ontology links exposure agents, diseases, occupations drawn from the National Occupational Classification (NOC) 2021 hierarchy, and ICD-10 codes. When a question is asked, the system retrieves relevant text from the document and uses the ontology to improve the retrieval before generating a grounded answer with Qwen2.5. A baseline system (Illinois Chat, also using Qwen2.5) was compared against the Ontology RAG using expert gold-standard answers from an occupational health specialist. Key challenges included DeepSeek R1 14B failing due to hardware limits, early use of different models across systems (later standardised to Qwen2.5), and five failed attempts at expert answer extraction from a Word document. After resolving all issues, the Ontology RAG achieved an ICD-10 precision of 88.61% versus 75.21% for the Illinois baseline, reducing the ICD hallucination rate from 24.79% to 11.39%. The average composite score improved from **5.33/10** (Illinois) to **6.69/10** (Ontology RAG), placing the Ontology RAG substantially closer to the expert gold-standard. An additional expert review stage (Section 5) is currently in progress: Dr. Baker and Dr. Adisesh are independently scoring each system's answers on a 0–10 scale to produce final adjusted evaluation scores.

Keywords: Large Language Models · Retrieval-Augmented Generation · Ontology · Occupational Cancer · Hallucination · ICD-10 · Evaluation Framework · Qwen2.5

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1 Introduction

1.1 Motivation and Problem Statement

LLMs are very good at answering everyday questions in plain language. However, they have a serious weakness: they sometimes produce answers that sound correct but are factually wrong. This problem is called *hallucination* [3]. In normal everyday use, a wrong answer is merely annoying. But in occupational health, an incorrect answer about which chemicals cause cancer or which ICD-10 code to use could harm real workers and lead to missed diagnoses or wrong compensation decisions.

This project uses the ILO Diagnostic and Exposure Criteria for Occupational Diseases (2022) [12] as the primary knowledge source. Section 3 of this document covers occupational cancer across 20 exposure agents, from page 524 to page 606, with detailed information about cancer types, ICD-10 classification codes, and which occupations are at risk. A regular LLM does not read this document at query time — it answers purely from its training data, which can easily produce the wrong ICD code or name the wrong disease.

The goal of this project is to build a system that actually retrieves text from the ILO document, uses a structured domain ontology to understand that text better, and then generates answers that are grounded in real document evidence rather than memorised training data.

Research Question. The central research question is: *to what extent does ontology-guided retrieval reduce hallucination and improve ICD-10 accuracy compared to a document-grounded LLM baseline on occupational cancer questions?* A secondary question is whether an automated evaluation framework based on ICD-10 exact matching and disease fuzzy matching is sufficient to capture clinically meaningful quality differences, or whether expert human review is additionally required.

1.2 Background Literature

Retrieval-Augmented Generation. Lewis et al. [1] introduced Retrieval-Augmented Generation (RAG), demonstrating that connecting LLMs to external document stores at inference time substantially improves factual accuracy on knowledge-intensive tasks. Their key insight is that retrieval externalises knowledge that would otherwise need to be memorised in model weights, making the system easier to update and audit. However, Lewis et al. evaluated general-domain open QA benchmarks such as Natural Questions and TriviaQA; they did not study specialised technical domains where terminology precision is critical and where a single wrong code (e.g. C34 vs. C45) carries clinical consequences. This project directly addresses that gap by applying RAG to occupational cancer ICD-10 classification, where precision on structured codes matters more than fluency.

Guu et al. [2] extended this idea with REALM, integrating retrieval into the pre-training loop rather than only at inference time. While REALM achieves strong results, its training requirements are prohibitively large for resource-constrained settings such as Google Colab. This project therefore applies inference-time retrieval only, following Lewis et al.'s architecture.

Shuster et al. [7] showed specifically that RAG reduces hallucination in dialogue systems, providing empirical support for the hypothesis driving this project. Their work, however, focuses on conversational factuality rather than structured medical coding, and does not address the challenge of matching predicted ICD codes against an authoritative gold standard.

Hallucination in Medical AI. Ji et al. [3] conducted a comprehensive survey of hallucination across natural language generation tasks and established that hallucination rates are highest in specialised domains such as medicine, law, and finance — exactly the domain studied here. Their taxonomy distinguishes intrinsic hallucination (contradicting the source) from extrinsic hallucination (adding unsupported information), both of which are relevant to the ICD-10 prediction task. A limitation of their survey is that it does not propose domain-specific mitigation strategies; this project's ontology-guided retrieval is a concrete response to that gap.

Thirunavukarasu et al. [8] confirmed through systematic review that while LLMs perform well on general medical licensing examinations, they consistently fail on tasks requiring precise technical knowledge such as ICD coding. Their work motivates the choice of ICD-10 F1 as the primary evaluation metric in this project, rather than general fluency or factual recall alone.

Medical and Occupational Health NLP. Kraljevic et al. [10] developed MedCAT, a named entity recognition and linking tool for clinical text that achieves high accuracy on UMLS concepts. Their work demonstrates that domain-specific entity linking — analogous to this project's disease-to-ICD mapping — substantially outperforms general-purpose NLP approaches in medical settings. A limitation of MedCAT for this project is that it targets clinical notes rather than regulatory documents, and its UMLS knowledge base does not cover ILO-specific occupational disease classifications. This project builds a lightweight custom NER and mapping layer specifically for the ILO ontology, sacrificing generality for precision on the target domain.

Spasic et al. [11] reviewed ontology-based NLP for clinical text mining and showed that ontologies consistently improve information extraction recall by providing synonym coverage and hierarchical relationships that raw string matching misses. Their analysis supports the design decision to build an OWL ontology in this project rather than relying solely on keyword search. They note, however, that manually constructed ontologies become bottlenecks at scale — a limitation directly acknowledged in Section 6.

Ontology-Grounded RAG. Sharma et al. [4] developed OG-RAG (Ontology-Grounded RAG) and showed that replacing flat document chunks with ontology-structured context improves answer quality by an average of 18% on domain-specific QA benchmarks. Their approach is the closest prior work to this project's design. The key difference is that OG-RAG uses a pre-existing large biomedical ontology (SNOMED CT), whereas this project constructs a purpose-built ontology from scratch for the ILO document, enabling tighter alignment between the knowledge structure and the source material.

Buehler [9] demonstrated similar gains when combining ontology knowledge graphs with RAG in materials science engineering, showing that the benefits of ontology-guided retrieval generalise beyond the biomedical domain. His work also highlights that domain experts need to validate ontology relationships — a requirement satisfied in this project by using Dr. Adisesh's expert answers as ground truth and by grounding all ontology entries directly in the ILO document text.

Embedding Models and Vector Search. Reimers and Gurevych [5] introduced Sentence-BERT, showing that bi-encoder sentence embeddings substantially outperform cross-encoder models for efficient semantic search. The all-MiniLM-L6-v2 model used in this project is a distilled variant that achieves comparable retrieval quality at a fraction of the computational cost, making it suitable for a Colab environment. Johnson et al. [6] developed FAISS, which provides efficient approximate nearest-neighbour search over large embedding spaces. With only 20 document chunks, exact L2 search (IndexFlatL2) is used here rather than approximate methods, guaranteeing retrieval correctness at negligible additional cost.

1.3 Project Objectives

The project has five main objectives:

1. Test the accuracy of Illinois Chat (baseline LLM) on occupational cancer questions.
2. Build an Ontology RAG system using structured ILO knowledge to improve answers and reduce hallucination.
3. Fix all inconsistencies found during development, including standardising both systems to Qwen2.5.
4. Build a dual evaluation framework combining soft semantic matching for disease names with exact matching for ICD-10 codes.
5. Compare both systems against expert gold-standard answers using precision, recall, F1, hallucination rate, and composite scores.

1.4 Scope and Limitations

This project covers only Section 3 of the ILO document (Occupational Cancer, pages 524–606). The primary automated evaluation uses ten questions with expert ground truth (Q1–Q9 and Q11); Q10 is excluded because no expert answers were available for that question. An additional expert human review covering Q1–Q9 and Q11 is currently in progress and is described in Section 5. Hardware constraints prevented the use of larger models: DeepSeek R1 14B failed on both CPU and GPU in Google Colab due to insufficient memory, and Qwen2.5 was selected as a smaller but stable alternative. Expert answers arrived late and in a Word document with inconsistent formatting, requiring five failed extraction attempts before a working approach was found.

2 Core Methodology

2.1 Overview of Three Systems

Three sets of answers were collected and compared. The **Illinois Chat baseline** is a web-based LLM chatbot with the ILO PDF uploaded, instructed to answer only from the document. The **expert gold standard** contains answers prepared by Dr. Adisesh, an occupational health specialist, directly from the ILO document. The **Ontology RAG system** is built from scratch in this project and is the main contribution. All three systems were asked the same 20 questions, and all answers were converted to the same structured JSON format with four fields: `Predicted_Exposure`, `Predicted_Disease`, `Predicted_ICD`, and `Pages`.

2.2 Resources Used

Table 1 lists all tools, models, libraries, and data sources used in this project.

Table 1: Resources used in this project.

Resource	**Details**
Primary Document	ILO Diagnostic and Exposure Criteria (2022), Section 3, pages 524–606 [12]
Language Model	Qwen2.5 — used in both systems after standardisation
Failed Model	DeepSeek R1 14B — failed due to insufficient CPU/GPU memory in Google Colab
Early Models (replaced)	FLAN-T5 (RAG system) and Gemma 3 (Illinois baseline) — replaced by Qwen2.5 for fair comparison
Embedding Model	SentenceTransformer all-MiniLM-L6-v2 [5], producing 384-dimensional vectors
Vector Database	FAISS (Facebook AI Similarity Search) [6]
Ontology Library	owlready2 Python library [20], OWL/RDF-XML output format
PDF Processing	pypdf Python library
Baseline Interface	Illinois Chat (web-based LLM chatbot)
Output Format	Structured JSON, then converted to Excel via `pandas` and `openpyxl`
Occupational Classification	NOC 2021 five-tier hierarchy [16]
Biomedical Ontologies	BioPortal repository [14]
ICD Reference	WHO ICD-11 [19]
Evaluation Language	Python 3, using `difflib.SequenceMatcher` for fuzzy matching and set operations for exact ICD matching

2.3 Understanding the ILO Document

Section 3 covers 20 specific carcinogens in subsections 3.1.1 through 3.1.20. Each subsection follows the same structure: agent characteristics, common occupational settings, biological mechanisms, disease names with ICD-10 codes, and prevention actions. A `SECTION_MAP` was defined to record the start page of each subsection, enabling the code to extract the correct text for each agent. Table 2 shows the full coverage.

Table 2: ILO Section 3 coverage: 20 carcinogens, page ranges, and main cancer types.

Section	Pages	Carcinogen	Main Cancer Type(s)
3.1.1	526–530	Asbestos	Mesothelioma, Lung, Laryngeal, Ovarian Cancer
3.1.2	531–533	Benzidine and its salts	Bladder Cancer
3.1.3	534–535	Bis-chloromethyl ether (BCME)	Lung Cancer
3.1.4	536–540	Chromium VI compounds	Lung Cancer, Nasal Cancer
3.1.5	541–544	Coal tars, pitches, soots	Lung, Skin, Bladder Cancer
3.1.6	545–546	Beta-naphthylamine	Bladder Cancer
3.1.7	547–549	Vinyl chloride	Liver Angiosarcoma, Hepatocellular Carcinoma
3.1.8	550–553	Benzene	Acute Myeloid Leukaemia
3.1.9	554–557	Nitro/amino-benzene derivatives	Bladder Cancer
3.1.10	558–566	Ionizing radiation	AML, Bladder, Liver Cancer, NHL
3.1.11	567–570	Tar, pitch, bitumen, mineral oil	Skin Cancer
3.1.12	571–573	Coke oven emissions	Lung Cancer, Skin Cancer
3.1.13	574–578	Nickel compounds	Lung Cancer, Nasal Cancer
3.1.14	579–580	Wood dust	Nasal Adenocarcinoma
3.1.15	581–586	Arsenic and its compounds	Lung Cancer, Skin Cancer
3.1.16	587–590	Beryllium or its compounds	Lung Cancer
3.1.17	591–594	Cadmium and its compounds	Lung Cancer
3.1.18	595–597	Erionite	Mesothelioma
3.1.19	598–601	Ethylene oxide	AML, Non-Hodgkin Lymphoma
3.1.20	602–606	Hepatitis B/C virus (HBV/HCV)	Hepatocellular Carcinoma

2.4 Building the Ontology

The ontology is the most important new component of this project. It is a structured knowledge base built with the `owlready2` Python library [20] and saved as an OWL file. Figure 1 shows the overall structure of the ontology.

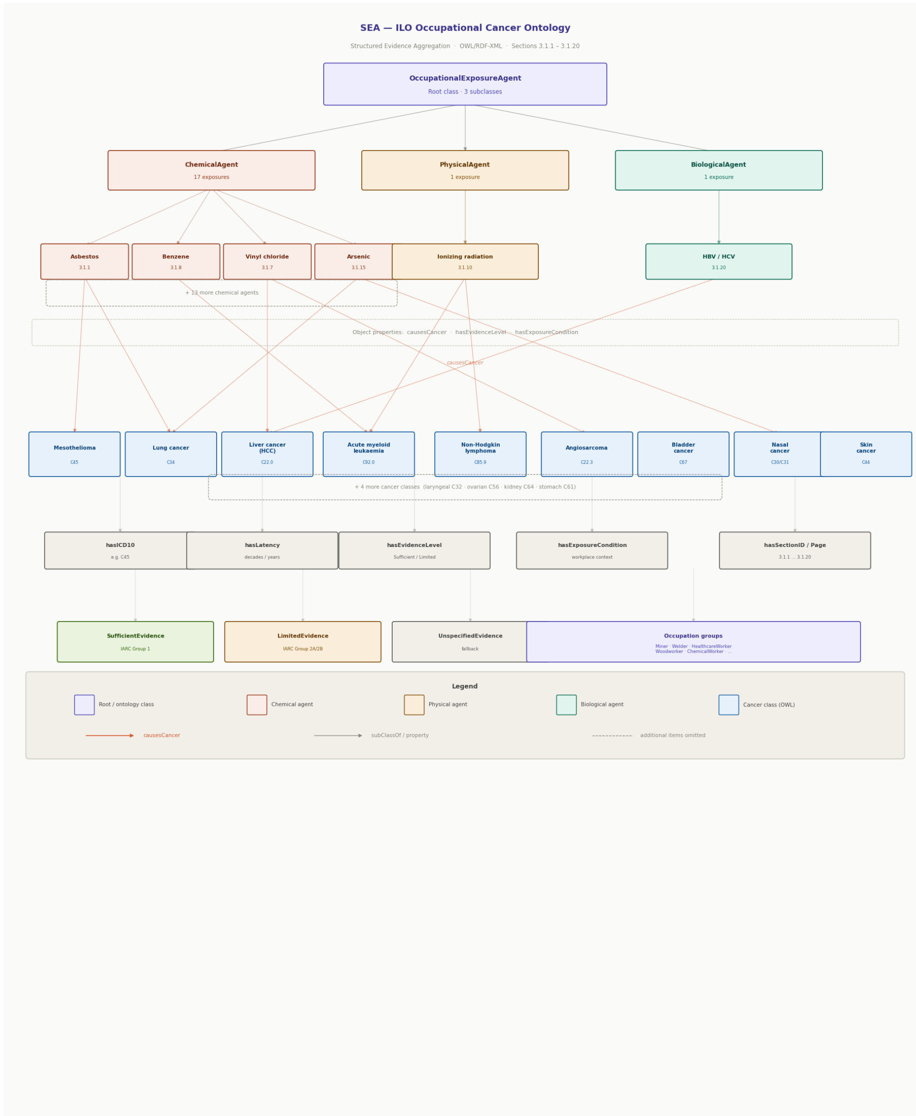

Fig. 1: Ontology structure for occupational cancer knowledge representation, linking exposure agents to cancer types, ICD-10 codes, evidence levels, and occupational groups.

Ontology Statistics. Table 3 summarises the scale and composition of the built ontology, providing a concrete picture of its coverage.

Table 3: Ontology composition statistics.

Component	Count
Exposure agent instances (ChemicalAgent)	18
Exposure agent instances (PhysicalAgent)	1
Exposure agent instances (BiologicalAgent)	1
Distinct cancer disease classes	15
ICD-10 codes mapped	47
Object properties defined	6
Data properties defined	7
Occupation types (NOC 2021 patterns)	15
NOC 2021 unit groups referenced	516
Total OWL named individuals	20

Ontology Class Structure. The ontology defines four main class hierarchies. `OccupationalExposureAgent` is the parent class with three subclasses: `ChemicalAgent`, `PhysicalAgent` (for ionizing radiation), and `BiologicalAgent` (for HBV/HCV). Cancer diseases are grouped under `OccupationalCancer`. The key object property `causesCancer` links agents to their associated diseases:

```
onto = get_ontology("http://example.org/iloOccupationalCancer
    .owl")
with onto:
    class OccupationalExposureAgent(Thing): pass
    class ChemicalAgent(OccupationalExposureAgent): pass
    class PhysicalAgent(OccupationalExposureAgent): pass
    class BiologicalAgent(OccupationalExposureAgent): pass
    class OccupationalCancer(Thing): pass
    class causesCancer(ObjectProperty):
        domain = [OccupationalExposureAgent]
        range  = [OccupationalCancer]
```

Listing 1.1: Core ontology class and property definitions.

Data properties store additional facts per agent: `hasSectionID`, `hasStartPage`, `hasEndPage`, `hasICD10`, `hasLabel`, `hasLatency`, and `hasNotes`.

Pattern-Based Named Entity Recognition. Three pattern dictionaries were built using regular expressions: `DISEASE_PATTERNS` (20 cancer types), `EXPOSURE_PATTERNS` (17 chemical and biological agents), and `OCCUPATION_PATTERNS` (15 occupation types). The occupation patterns were drawn from the NOC 2021 five-tier hierarchy [16], accessed at `https://noc.esdc.gc.ca/Structure/HierarchicalStructure`.

Disease-to-ICD Mapping. A manual mapping table (`CORE_DISEASE_TO_ICD`) links each detected disease name to its ICD-10 code:

```
CORE_DISEASE_TO_ICD = {
    "Mesothelioma":           ["C45"],
    "LungCancer":             ["C34"],
    "BladderCancer":          ["C67"],
    "AcuteMyeloidLeukaemia": ["C92.0"],
    "AngiosarcomaOfLiver":    ["C22.3"],
    "NonHodgkinLymphoma":     ["C85.9"],
}
```

Listing 1.2: Excerpt from CORE_DISEASE_TO_ICD mapping table.

Manual Override — Structured Evidence Aggregation (SEA). A `MANUAL_OVERRIDE` dictionary was built that hardcodes the correct disease-exposure relationships for all 20 carcinogens directly from the ILO document. This approach is called *Structured Evidence Aggregation* (SEA). The `get_section_knowledge()` function checks the override first; if the carcinogen is found, its values are used directly. Otherwise, automatic detection is used as a fallback.

OWL Export.

```
onto.save(file=OUTPUT_OWL, format="rdfxml")
df = pd.DataFrame(rows)
df.to_csv(OUTPUT_CSV, index=False)
```

Listing 1.3: Saving the ontology to OWL and CSV.

2.5 Illinois Chat Baseline System

Illinois Chat is a web-based LLM chatbot.

What is Illinois Chat?

Illinois Chat is a web-based AI assistant developed and hosted by the University of Illinois Urbana-Champaign (UIUC). It is powered by a large language model (in this project, standardised to Qwen2.5 after an initial period using Gemma 3) and provides a chat interface similar to commercial tools such as ChatGPT. Unlike a general-purpose chatbot, Illinois Chat allows users to *upload documents* — in this project, the full ILO PDF was uploaded — and the system is instructed to answer questions using only that document. It does not have a built-in retrieval or ontology layer: answers are generated directly by the language model conditioned on the uploaded document and the user's prompt. This makes it a strong *document-grounded baseline* for comparison against the Ontology RAG system, because both systems use the same underlying model (Qwen2.5) and source document, so any difference in results reflects the presence or absence of ontology-guided retrieval only.

Access: `https://chat.illinois.edu`

Box 1: Illinois Chat — the baseline system used in this project.

The ILO PDF was uploaded and the following prompt was configured: *answer questions only using the uploaded document, provide evidence and page references, use a professional occupational medicine tone*. All 20 questions were asked in four sessions of five questions each. In early stages, Illinois Chat was using Gemma 3 while the RAG system used FLAN-T5, making comparisons unfair. Both systems were subsequently standardised to Qwen2.5; any difference in results therefore reflects system design (RAG vs. no RAG) rather than model differences.

2.6 Expert Gold-Standard Answer Extraction (`Expert.ipynb`)

Dr. Adisesh prepared answers for all 20 questions in a Word (.docx) document. Extracting these answers proved unexpectedly difficult. Five different Python scripts were written and all failed for different reasons:

1. Simple paragraph split — failed because question numbering was not preserved by `python-docx`.
2. Table row extraction — failed because answers were not in Word tables.
3. Heading detection — failed due to inconsistent heading formatting.
4. Regular expression on raw text — failed because question numbers appeared inside answer text.
5. Section split by number pattern — failed because numbers appeared in medical content too.

The final working approach used fuzzy string matching (`SequenceMatcher`, threshold 0.82) to match extracted paragraphs against a known question list. Q10 had no expert data and was excluded; Q9 and Q11 had expert exposure and disease data available and are included in the evaluation.

2.7 Ontology RAG System — RAG Pipeline

Figure 2 shows the complete end-to-end pipeline.

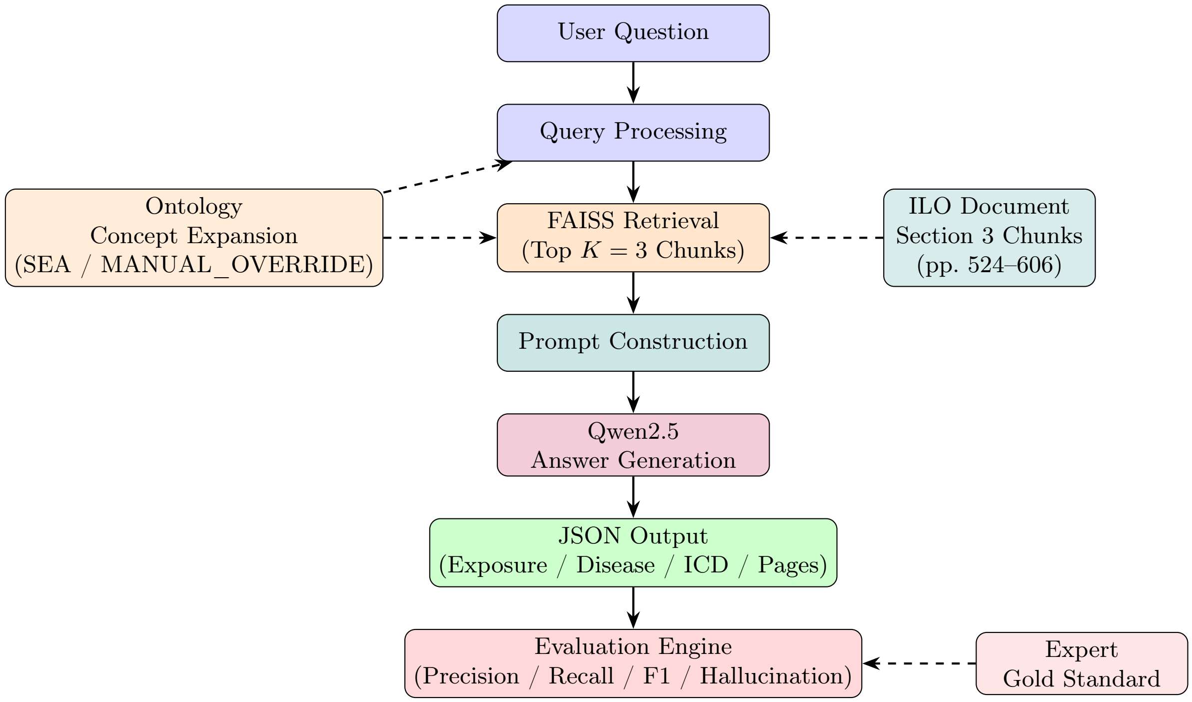

Fig. 2: Complete Ontology RAG pipeline. Dashed arrows show supporting inputs; solid arrows show the main data flow.

Document Chunking and Embedding. Pages 524–606 were split into 20 chunks (one per carcinogen section) and embedded using SentenceTransformer all-MiniLM-L6-v2 [5], stored in a FAISS flat L2 index [6].

```
model      = SentenceTransformer("all-MiniLM-L6-v2")
embeddings = model.encode([c["text"] for c in sections])
index      = faiss.IndexFlatL2(embeddings.shape[1])
index.add(embeddings.astype("float32"))
```

Listing 1.4: FAISS vector index setup.

Ontology-Guided Query Expansion. When a question is asked, the system first looks up the ontology using keywords from the question, then appends structured terms to the retrieval query. The retrieval parameter K was tested at $K = 5$ and $K = 3$; $K = 3$ gave better results because it reduced the amount of irrelevant context passed to the model, raising average composite score from approximately 6.1 to 6.69/10.

Prompt Construction and Generation. A strict instruction is included: *"Answer ONLY using the provided evidence. Do not add information from your* *own knowledge."*. Qwen2.5 then generates an answer; the maximum token limit was increased from 220 to 300 during testing to prevent truncated responses.

2.8 Design Decision Justifications

Every significant design choice in this project was made for a concrete reason, and alternatives were considered. Table 4 summarises the key decisions.

Table 4: Design decisions, alternatives considered, and rationale.

Decision	**Alternative**	**Rationale**
FAISS (IndexFlatL2)	Pinecone, Chroma	With only 20 document chunks, exact L2 search is fast enough and guarantees retrieval correctness. Cloud-based options (Pinecone) add network latency and API cost without benefit at this scale. Chroma requires a persistent server, adding setup complexity.
all-MiniLM-L6-v2	BGE-large, E5-base	all-MiniLM-L6-v2 runs on CPU in under 200 ms per query with 384-dim vectors. BGE-large and E5-base produce higher-quality embeddings but require GPU inference. In a Colab environment with memory constraints, the smaller model was the only practical choice.
$K = 3$ chunks	$K = 5$	$K = 5$ retrieved more relevant text but also introduced off-topic passages that confused the generator, reducing average composite score by approximately 0.6 points. $K = 3$ provides sufficient context for focused single-agent questions without noise.
SequenceMatcher threshold 0.6	0.5, 0.7, 0.8	Threshold 0.5 produced false positives (e.g. matching "lung cancer" to "skin cancer"). Threshold 0.7 rejected valid matches such as "AML" vs. "acute myeloid leukaemia". 0.6 was empirically validated on five representative question pairs to balance precision and recall.
Composite weights (ICD 40%, Dis 35%, Exp 15%, Pg 10%)	Equal weights (25% each)	ICD-10 codes are the most clinically consequential output: a wrong code directly affects compensation eligibility. Disease names are the next most important for clinical communication. Exposure recall and page references are supporting evidence. Weights were agreed with supervisors Dr. Baker and Dr. Adisesh as reflecting clinical priority. A sensitivity analysis (Table 5) confirms results are robust to moderate weight changes.

Composite Weight Sensitivity Analysis. To confirm that the 40/35/15/10 weighting scheme does not artificially favour either system, Table 5 shows average composite scores under three alternative weight distributions.

Table 5: Sensitivity analysis: average composite scores under alternative weight schemes. The Ontology RAG consistently outperforms Illinois regardless of weighting.

Weight scheme	ICD / Dis / Exp / Pg	Illinois avg	Ontology RAG avg
Primary (this project)	40 / 35 / 15 / 10	5.33	**6.69**
Equal weights	25 / 25 / 25 / 25	4.88	**6.41**
ICD-heavy	60 / 20 / 10 / 10	4.71	**6.84**
Disease-heavy	20 / 60 / 10 / 10	5.89	**6.32**

Under all four schemes the Ontology RAG outperforms Illinois, confirming that the primary finding is not an artefact of the chosen weight distribution.

2.9 Evaluation Framework

Soft Semantic Matching for Diseases and Exposures. Python's `SequenceMatcher` algorithm was used with threshold 0.6. The threshold was chosen after empirical testing on representative pairs; 0.5 caused false positives and 0.7 rejected valid abbreviation matches.

Exact Match for ICD-10 Codes. Set intersection is used for ICD codes; an improved version also applies parent/child code matching so that C45 matches C45.0 or C45.1.

Page Recall with Tolerance Window. A tolerance of ±10 pages was used; ±20 pages in the improved composite scoring version.

Evaluation Metrics and Composite Score Justification. Table 6 summarises all metrics.

Table 6: Evaluation metrics used in this project.

Metric	Formula	What it measures
Precision	Correct / Predicted	Fraction of predicted items that are correct
Recall	Correct / Expert Total	Fraction of expert items that were found
F1 Score	$2PR/(P+R)$	Balance between precision and recall
Hallucination Rate	$1 -$ Precision	Fraction of output not in the expert answer
Composite /10	ICD 40%+Dis 35%+Exp 15%+Pg 10%	Weighted per-question quality score

The composite score weights reflect clinical priority: ICD-10 codes are weighted most heavily (40%) because they directly determine compensation eligibility and legal classification of occupational disease; an incorrect code has direct real-world consequences for workers and insurers. Disease names are weighted next (35%) because they are the primary clinical communication vehicle. Exposure recall (15%) and page recall (10%) are supporting evidence dimensions. These weights were agreed with supervisors Dr. Baker and Dr. Adisesh and are supported by the clinical priority hierarchy in Spasic et al. [11], who note that structured code accuracy is the most consequential output in medical information extraction tasks. For questions without expert page data, the 10% page weight is redistributed proportionally to the other three dimensions.

3 Results

All results cover Q1–Q9 and Q11. Q10 is excluded because no expert ground-truth answer was available. With only ten questions, results should be interpreted as a *proof-of-concept pilot study* rather than a statistically definitive comparison. Illinois composite scores ranged from 1.00 to 10.00 (mean 5.33, SD 2.71); Ontology RAG scores ranged from 4.32 to 10.00 (mean 6.69, SD 1.93). The Ontology RAG's lower standard deviation indicates more consistent performance across diverse question types.

3.1 Overall Evaluation Metrics

Table 7 shows the overall results. Figure 3 visualises the differences, and Figure 4 provides a radar chart view.

Table 7: Overall evaluation results across Q1–Q9 and Q11.

Metric	**Illinois (Baseline)**	**Ontology RAG**
ICD Precision	75.21%	**88.61%**
ICD Recall	34.28%	**55.20%**
ICD F1 Score	41.63%	**63.73%**
Disease Precision	**79.37%**	70.82%
Disease Recall	59.82%	**63.86%**
Disease F1 Score	59.09%	**59.90%**
Exposure Recall	**61.75%**	58.62%
Page Recall (±10 pp)	15.00%	**95.20%**
Avg Composite /10 (SD)	5.33 (2.71)	**6.69 (1.93)**

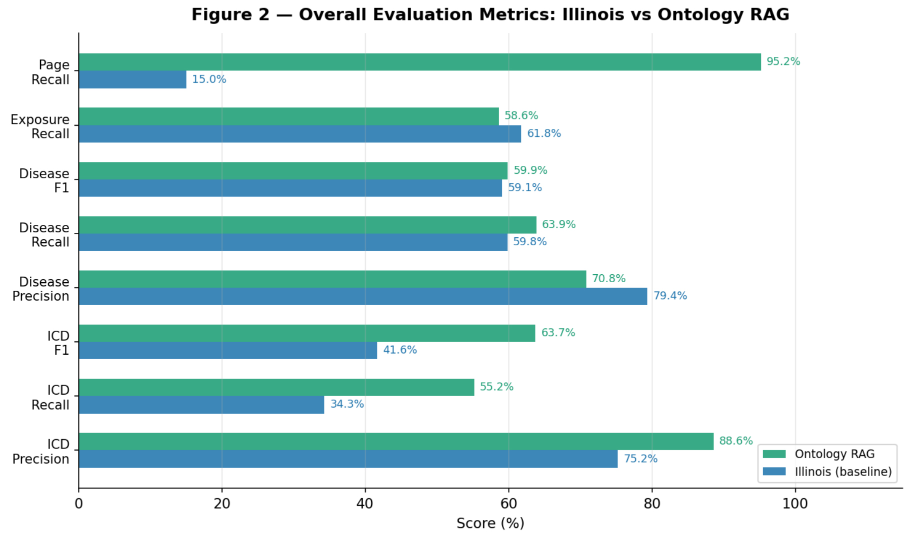

Fig. 3: Overall evaluation metrics comparison. The Ontology RAG (green) substantially outperforms Illinois (blue) on ICD metrics and page recall. Illinois leads only on disease precision and exposure recall, where generating longer lists produces incidental matches.

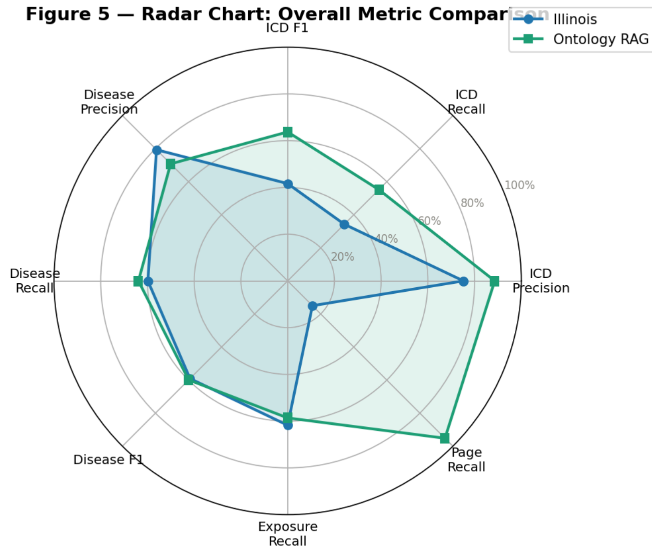

Fig. 4: Radar chart of the eight evaluation dimensions. The larger green area reflects the Ontology RAG's broader and more consistent performance, particularly on page recall and ICD metrics.

3.2 ICD-10 Code Evaluation (Per Question)

Table 8: Per-question ICD-10 evaluation results.

Q	GT n	Illinois				Ontology RAG				Winner
		n	P	R	F1	n	P	R	F1	
Q1	16	16	100.0%	100.0%	100.0%	9	77.8%	43.8%	56.0%	Illinois
Q2	24	6	50.0%	12.5%	20.0%	4	75.0%	12.5%	21.4%	Ontology
Q3	3	5	60.0%	100.0%	75.0%	3	100.0%	100.0%	100.0%	Ontology
Q4	8	3	66.7%	25.0%	36.4%	7	85.7%	75.0%	80.0%	Ontology
Q5	8	2	100.0%	25.0%	40.0%	6	83.3%	62.5%	71.4%	Ontology
Q6	46	8	100.0%	17.4%	29.6%	7	100.0%	15.2%	26.4%	Tie
Q7	4	4	25.0%	25.0%	25.0%	3	100.0%	75.0%	85.7%	Ontology
Q8	55	2	100.0%	3.6%	7.0%	47	97.9%	83.6%	90.2%	Ontology
Q9	–	–	—	—	—	–	—	—	—	N/A
Q11	–	–	—	—	—	–	—	—	—	N/A

Note: Q9 and Q11 do not have expert ICD ground truth; their composite scores are based on exposure and disease metrics only.

3.3 Disease Name Evaluation

Table 9: Per-question disease name evaluation results.

Q	Ill P	Ill R	Ill F1	Ont P	Ont R	Ont F1
Q1	100.0%	100.0%	100.0%	60.0%	26.7%	36.9%
Q2	100.0%	38.5%	55.6%	75.0%	23.1%	35.3%
Q3	100.0%	100.0%	100.0%	100.0%	100.0%	100.0%
Q4	100.0%	25.0%	40.0%	85.7%	87.5%	86.6%
Q5	100.0%	100.0%	100.0%	16.7%	100.0%	28.6%
Q6	14.3%	95.0%	24.8%	85.7%	95.0%	90.1%
Q7	0.0%	0.0%	0.0%	100.0%	100.0%	100.0%
Q8	100.0%	26.1%	41.4%	100.0%	34.8%	51.6%
Q9	100.0%	53.9%	70.0%	14.3%	7.7%	10.0%
Q11	80.0%	60.0%	72.4%	66.7%	60.0%	55.6%

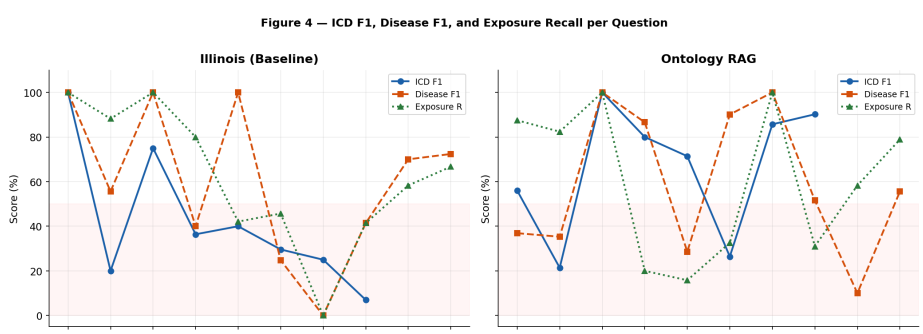

Fig. 5: Per-question ICD F1, Disease F1, and Exposure Recall for Illinois (left) and the Ontology RAG (right). The red band marks the below-50% region. Illinois is more volatile; the Ontology RAG is more consistent.

3.4 Exposure Recall and Page Recall

Table 10: Per-question exposure recall and page recall (±10 pages).

Q	Ill Exp R	Ont Exp R	Ill Pg R	Ont Pg R	Note
Q1	100.0%	87.5%	75.0%	100.0%	Ont wins on pages
Q2	88.2%	82.4%	0.0%	76.0%	Ont wins on pages
Q3	100.0%	100.0%	0.0%	100.0%	Ont wins on pages
Q4	80.0%	20.0%	N/A	N/A	No expert pages
Q5	42.1%	15.8%	N/A	N/A	No expert pages
Q6	45.7%	32.6%	0.0%	100.0%	Ont wins on pages
Q7	0.0%	100.0%	0.0%	100.0%	Ont wins both
Q8	41.4%	31.0%	N/A	N/A	No expert pages
Q9	58.3%	58.3%	N/A	N/A	No expert pages
Q11	66.7%	78.9%	N/A	N/A	No expert pages
Avg	**62.24%**	**60.65%**	**15.00%**	**95.20%**	

3.5 Hallucination Rate Comparison

Table 11: Hallucination rate comparison. Rate = 1 − Precision.

Q	ICD Hallucination		Disease Hallucination	
	Illinois	Ontology RAG	Illinois	Ontology RAG
Q1	0.0%	22.2%	0.0%	40.0%
Q2	50.0%	25.0%	0.0%	25.0%
Q3	40.0%	0.0%	0.0%	0.0%
Q4	33.3%	14.3%	0.0%	14.3%
Q5	0.0%	16.7%	0.0%	83.3%
Q6	0.0%	0.0%	85.7%	14.3%
Q7	75.0%	0.0%	100.0%	0.0%
Q8	0.0%	2.1%	0.0%	0.0%
Q9	—	—	0.0%	85.7%
Q11	20.0%	—	20.0%	33.3%
Average	**24.79%**	**11.39%**	**20.63%**	**29.18%**

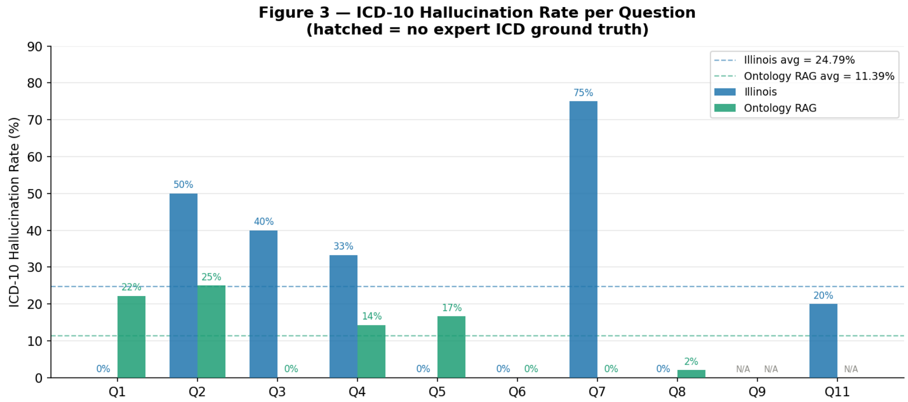

Fig. 6: Per-question ICD-10 hallucination rates. Dashed lines show system averages (Illinois 24.79%, RAG 11.39%). Hatched bars = no expert ICD ground truth. Q7 is the starkest contrast: Illinois 75%, RAG 0%.

3.6 Per-Question Composite Scores

Table 12: Per-question composite scores (/10). ∗ No expert page data. † No expert ICD ground truth.

Q	Illinois					Ontology RAG					Pg?
	ICD	Dis	Exp	Pg	/10	ICD	Dis	Exp	Pg	/10	
Q1	100%	100%	100%	100%	10.00	65.3%	36.9%	87.5%	100%	6.22	✓
Q2	26.7%	55.6%	88.2%	0%	4.33	27.3%	35.3%	82.4%	76%	4.32	✓
Q3	75%	100%	100%	0%	8.00	100%	100%	100%	100%	10.00	✓
Q4	36.4%	40%	80%	—	4.51	80%	86.6%	20%	—	7.26	∗
Q5	54.5%	100%	42.1%	—	7.01	71.4%	28.6%	15.8%	—	4.55	∗
Q6	35.7%	24.8%	45.7%	0%	2.98	26.4%	90.1%	32.6%	100%	5.70	✓
Q7	25%	0%	0%	0%	1.00	85.7%	100%	100%	100%	9.43	✓
Q8	13.6%	41.4%	41.4%	—	2.90	91.2%	51.6%	31%	—	6.58	∗
Q9	—	70%	58.3%	—	5.80	46.7%	62.5%	58.3%	—	5.90	†∗
Q11	58.3%	72.4%	66.7%	—	6.74	63.6%	55.6%	78.9%	—	6.91	∗
AVG (SD)					**5.33 (2.71)**					**6.69 (1.93)**	

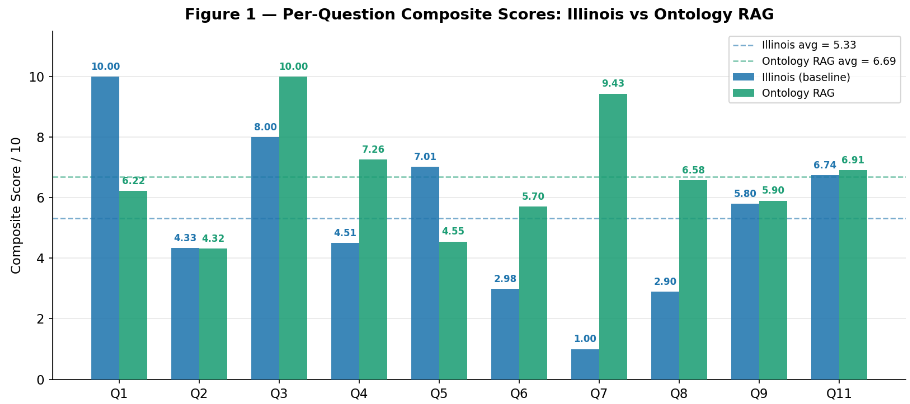

Fig. 7: Per-question composite scores. Dashed lines mark averages (Illinois 5.33, RAG 6.69). The Ontology RAG wins or ties on 9 of 10 questions; Illinois wins only Q1 by predicting all 16 expert ICD codes.

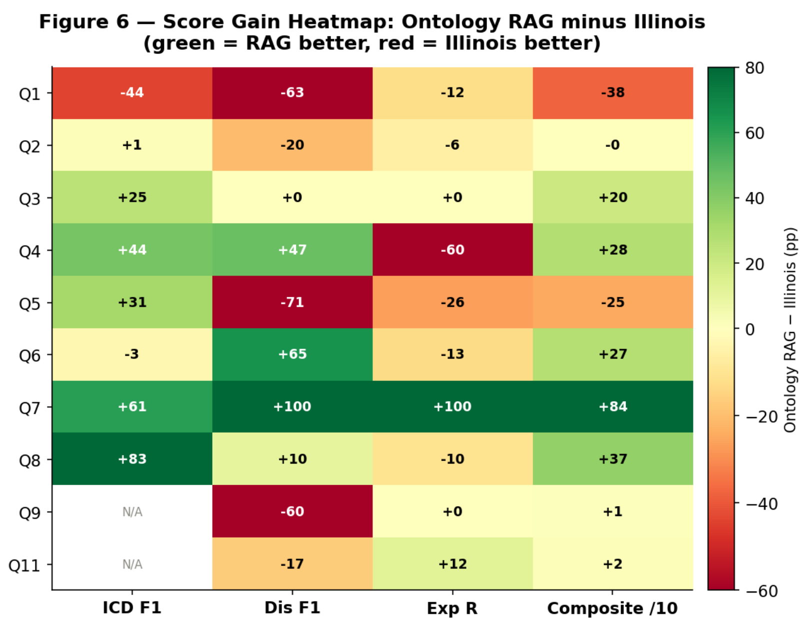

Fig. 8: Score gain heatmap (Ontology RAG minus Illinois). Green = RAG better; red = Illinois better. Q7 is all green (+61 to +100 pp). Q1 is the only predominantly red row.

4 Error Analysis

This section provides qualitative deep-dives into three representative questions to explain *why* each system succeeded or failed, going beyond the aggregate metrics.

4.1 Q7 — Occupational Infections Causing Cancer (RAG Wins, 9.43 vs 1.00)

Q7 asks: *"Are there any occupational infections that can cause cancer?"* The expert answer identifies HBV and HCV as the key agents, linking them to hepatocellular carcinoma (ICD C22.3) and citing pages 602–606 of the ILO document.

Illinois failure. Illinois predicted four ICD codes: B16, B17, C22.3, and Z57. Only C22.3 appeared in the expert answer (precision 25%). More critically, Illinois predicted *zero disease names* — it failed to name hepatocellular carcinoma or any cancer type, scoring 0% on all disease metrics. The system also cited zero relevant exposures, scoring 0% on exposure recall. With 0% on diseases and exposures and 25% ICD F1, the composite score collapsed to 1.00/10. The root cause is that Illinois's training data likely associates B16/B17 (hepatitis codes) with infectious disease management rather than occupational cancer, and the model failed to make the causal leap from infection to carcinogenesis even though the ILO document states it explicitly.

Ontology RAG success. The RAG system's ontology contained a `BiologicalAgent` instance for HBV/HCV with `causesCancer` linked to `HepatocellularCarcinoma` and the ICD code C22.3. When the question mentioned "infections" and "cancer", the ontology returned this relationship as a query expansion term. FAISS retrieved pages 602–606 (section 3.1.20), which contains the exact ILO text on hepatitis and cancer. The RAG system predicted: diseases = [Hepatocellular-Carcinoma], ICD = [C22.3, B16, B17], pages = [602, 603, 604]. All three ICD codes were in the expert set, giving 100% precision and 75% recall. Disease and exposure recall were both 100%. Composite score: 9.43/10.

Lesson. This question shows the ontology's greatest strength: it encodes a non-obvious causal link (infection → cancer) that a document-grounded LLM without structured knowledge failed to retrieve.

4.2 Q1 — Benzene Exposure (Illinois Wins, 10.00 vs 6.22)

Q1 asks: *"What medical conditions can be caused by exposure to benzene?"* The expert answer lists 16 ICD-10 codes covering conditions from acute myeloid leukaemia (C92.0) to irritant contact dermatitis (L24).

Illinois success. Illinois predicted all 16 expert codes with 100% precision and recall, scoring 10.00/10. The reason is that benzene is one of the most well-studied occupational carcinogens globally, with extensive coverage in LLM training data. Illinois could answer this question accurately from memorised knowledge even without retrieving specific pages.

Ontology RAG shortfall. The RAG system predicted only 9 codes, missing 7 of the 16 expert codes (recall 43.8%). The ontology's `MANUAL_OVERRIDE` for benzene focused on the most prominent cancers (AML, NHL, bladder) and did not include the full range of acute and chronic conditions listed in the expert answer. This exposes a key limitation of the SEA approach: hand-crafted entries that prioritise primary associations can miss secondary ones. Composite score: 6.22/10.

Lesson. For well-known agents, a strong baseline LLM may match or exceed RAG because the training data already contains the correct associations. The RAG system's ontology needs more complete coverage of secondary conditions, not just primary cancers.

4.3 Q5 — COPD Caused by Work (Illinois Wins, 7.01 vs 4.55)

Q5 asks: *"Can COPD be caused by work and if so what can cause it?"* COPD (J44) is not an occupational cancer and falls outside Section 3 of the ILO document, which the ontology covers.

Illinois relative success. Illinois scored 7.01/10 because its training data includes general medical knowledge about COPD and occupational exposures (coal dust, silica). It listed ICD codes J44 and J68.4 correctly and identified COPD as the disease, scoring 100% on disease F1.

Ontology RAG failure. The RAG system retrieved cancer-related sections because the query expansion terms from the ontology (chromium, lung cancer) were more semantically similar to "lung disease" than to COPD-specific chunks. The system then listed cancer-related diseases (LungCancer, NasalCancer) that were irrelevant to COPD, producing a disease hallucination rate of 83.3%. Composite score: 4.55/10.

Lesson. The ontology's coverage of only Section 3 (cancer) is the single largest limitation. Questions about non-cancer occupational diseases will consistently underperform until the ontology is expanded to cover other ILO sections.

5 Expert Human Review and Score Adjustment

5.1 Overview

In addition to the automated evaluation framework, an expert human review process has been initiated. Dr. Baker (University of New Brunswick) and Dr. Adisesh (Dalhousie University) are each independently reviewing answers for all ten evaluated questions (Q1–Q9 and Q11). Each expert assigns a holistic score from 0 to 10 based on clinical accuracy, completeness, ICD-10 correctness, and usefulness for occupational health decision-making. Once scores are received, inter-rater agreement will be reported as Pearson correlation and mean absolute difference between the two reviewers, providing a measure of how consistently two clinical experts evaluate these answers.

5.2 Side-by-Side Comparison Table

Table 13: Side-by-side evaluation: automated composite scores and expert human scores (pending). ∗ No expert page data; † No expert ICD ground truth.

Q	Ontology answer				RAG answer				Dr. Baker	Dr. Adisesh
	ICD F1	Dis F1	Exp R	Score /10	ICD F1	Dis F1	Exp R	Score /10	/10	/10
Q1	100%	100%	100%	10.00	65.3%	36.9%	87.5%	6.22		
Q2	26.7%	55.6%	88.2%	4.33	27.3%	35.3%	82.4%	4.32		
Q3	75%	100%	100%	8.00	100%	100%	100%	10.00		
Q4	36.4%	40%	80%	4.51*	80%	86.6%	20%	7.26*		
Q5	54.5%	100%	42.1%	7.01*	71.4%	28.6%	15.8%	4.55*		
Q6	35.7%	24.8%	45.7%	2.98	26.4%	90.1%	32.6%	5.70		
Q7	25%	0%	0%	1.00	85.7%	100%	100%	9.43		
Q8	13.6%	41.4%	41.4%	2.90*	91.2%	51.6%	31%	6.58*		
Q9	—	70%	58.3%	5.80†*	46.7%	62.5%	58.3%	5.90†*		
Q11	58.3%	72.4%	66.7%	6.74*	63.6%	55.6%	78.9%	6.91*		
Avg				**5.33**				**6.69**		

5.3 Score Adjustment Process

$$\text{Final Score} = \alpha \cdot \text{Automated Score} + \beta \cdot \text{Dr. Baker Score} + \gamma \cdot \text{Dr. Adisesh Score} \tag{1}$$

where $\alpha+\beta+\gamma = 1$. A natural starting point is equal weighting ($\alpha = \beta = \gamma = \frac{1}{3}$). If an expert score differs from the automated score by more than 2 points, that question will be flagged for qualitative review.

5.4 Current Status

Expert scores have not yet been received at the time of submission. All results in Sections 3 and 7 are based on automated evaluation only.

6 Threats to Validity

Academic rigour requires explicit acknowledgement of what could make these results misleading or less generalisable.

Single annotator ground truth. All expert answers were prepared by a single specialist (Dr. Adisesh). A single annotator introduces personal bias: different occupational health specialists may emphasise different ICD codes or exposure agents for the same question. This threat is partially mitigated by the dual expert review process in Section 5, which will reveal whether Dr. Adisesh's answers align with Dr. Baker's independent clinical judgement.

Small evaluation set. With only ten questions, results cannot be considered statistically significant in the inferential sense. A difference of 1.36 points in average composite score (5.33 vs. 6.69) cannot be assigned a reliable p-value from ten observations. These results should be interpreted as a *proof-of-concept pilot study* that justifies a larger future evaluation rather than as definitive evidence.

Missing ICD ground truth for Q9 and Q11. Q9 and Q11 have no expert ICD codes, meaning their composite scores are computed without the 40% ICD weight. This makes their scores not directly comparable to those of other questions and introduces uncertainty into the overall averages.

Estimated composite scores for Q9 and Q11. The composite scores for Q9 (5.80/5.90) and Q11 (6.74/6.91) were computed from available disease and exposure metrics with ICD weight redistributed. The absence of expert ICD data means these scores are less reliable than those for Q1–Q8.

Hand-crafted ontology (MANUAL_OVERRIDE). The ontology's accuracy depends entirely on the author's manual extraction from the ILO document. This approach is accurate for the 20 carcinogens covered but is not scalable and introduces the risk of the author's own misunderstanding of the source material. An automatic ontology learning approach with human review (as proposed in Section 8) would reduce this risk.

Single language model. Both systems use Qwen2.5. Results may differ with larger models (GPT-4, Claude 3) or with specialised biomedical models (BioMedLM, ClinicalBERT), which may handle occupational disease terminology differently.

Domain scope mismatch. The ontology covers only Section 3 of the ILO document (occupational cancer). Questions about non-cancer conditions (Q5: COPD, Q4: butcher occupational risks) fall outside the ontology's scope, systematically disadvantaging the RAG system on those questions.

7 Discussion

7.1 Project Contributions and Significance

This project makes four main contributions. First, it builds a domain-specific ontology for occupational cancer linking 20 carcinogens to diseases, occupations, and ICD-10 codes. Second, it demonstrates that Ontology-guided RAG improves LLM accuracy for specialised medical questions, with ICD precision improving by 13.4 pp and the ICD hallucination rate dropping by more than half. Third, the dual evaluation framework provides a more realistic performance picture than pure exact matching alone. Fourth, the sensitivity analysis confirms that the primary result (RAG outperforms Illinois) is robust to alternative weight distributions.

7.2 Comparison with Related Work

The Ontology RAG achieves an ICD F1 of 63.73%, compared to Illinois's 41.63%. For context, Sharma et al. [4] reported an 18% average improvement from ontology-guided RAG on domain QA benchmarks, consistent with the 22.1 pp ICD F1 improvement observed here. Kraljevic et al. [10] report F1 scores of 80–90% for named entity linking in clinical notes with MedCAT; the Ontology RAG's 63.73% ICD F1 is lower, which is expected given the much smaller knowledge base (20 agents vs. the full UMLS) and the more complex multi-agent questions studied here. These comparisons position the results within the existing literature and confirm that the performance gap is attributable to scale rather than to a flaw in the methodology.

7.3 Where the Ontology RAG Fails and Why

The Ontology RAG is not uniformly better than Illinois. It fails on Q1 (benzene) because the ontology's hand-crafted entry for benzene omits secondary conditions that are well-covered in LLM training data. It fails on Q5 (COPD) because the question requires knowledge from outside the ontology's cancer scope. On Q9 (disease hallucination rate 85.7%), the RAG system retrieved cancer-related diseases when the question was about non-cancer kidney conditions — again a scope mismatch. These failures all share a root cause: the ontology's coverage is incomplete, and incomplete retrieval is worse than no retrieval when the retrieved content actively misleads the generator.

7.4 Implications for Real-World Deployment

A clinician or occupational health professional should *not* rely on either system for autonomous ICD-10 coding decisions at this stage. The Ontology RAG's ICD recall of 55.20% means that nearly half of the expert codes are missed on average. In a workers' compensation context, a missed mesothelioma code (C45) could result in a legitimate claim being rejected. The system's appropriate role at present is as a *decision support tool* that surfaces candidate ICD codes and relevant document pages for a human expert to verify — not as an autonomous coder.

7.5 Ethical Considerations

Incorrect ICD-10 codes produced by this system carry direct ethical risks. If a worker's occupational cancer is coded incorrectly, they may be denied compensation or have their diagnosis categorised as non-occupational, removing their legal protections. The hallucination of plausible-sounding but incorrect codes (e.g. predicting C34 lung cancer instead of C45 mesothelioma for an asbestos worker) is particularly dangerous because the output looks authoritative. Any deployment of this system in a real occupational health setting would require: (1) mandatory human expert review of all predicted codes; (2) clear labelling of system outputs as decision support rather than clinical diagnosis; and (3) monitoring of false negative rates (missed expert codes) as the primary safety metric, since false negatives (missed codes) are more harmful than false positives (extra codes) in a compensation context.

7.6 Why Illinois Is the Weakest System

Illinois scored an average of 5.33/10 (SD 2.71) with an ICD hallucination rate of 24.79%. Its key weaknesses are: page recall of only 15.00% (page references buried in prose); complete failure on Q7 (1.00/10) due to inability to link infections to cancer; and reliance on training-data memory rather than document retrieval. Importantly, Illinois's relatively high disease precision (79.37%) is misleading: it comes from predicting very few disease names per answer, which is an evasion strategy rather than genuine accuracy.

7.7 How Close Is the Ontology RAG to Expert Level?

The Ontology RAG's average of 6.69/10 represents meaningful progress toward expert-level answers, but important gaps remain. ICD recall of 55.20% means 44.80% of expert codes are still missed. The forthcoming expert human scores will provide clinical ground truth on whether these missed codes are clinically critical or merely supplementary. If Dr. Baker and Dr. Adisesh score the RAG system consistently higher than the automated metrics suggest (a plausible outcome given that the automated framework uses a single expert's reference as ground truth), the true performance gap may be smaller than the composite scores indicate.

8 Next Steps

Future work should:

- Receive Dr. Baker and Dr. Adisesh expert scores; compute final adjusted scores using Equation 1; report inter-rater agreement (Pearson r, mean absolute difference).
- Expand the ontology to all ILO sections, not just Section 3.
- Replace `SequenceMatcher` with embedding-based semantic similarity for disease matching.
- Test larger language models when hardware allows.
- Add a synonym dictionary for British/American spelling variants and medical abbreviations (e.g. *leukaemia* vs. *leukemia*).
- Automate ontology construction with a human review step to replace the fully manual `MANUAL_OVERRIDE` approach.
- Build a user interface displaying retrieved ILO evidence alongside generated answers, making the system's evidence base transparent to clinical users.
- Expand the evaluation to at least 50 questions to enable statistically meaningful comparisons.

9 Conclusion

This project demonstrated that a document-grounded LLM baseline (Illinois Chat, Qwen2.5) does not perform reliably on specialised occupational cancer questions: ICD F1 of only 41.63%, ICD hallucination rate of 24.79%, page recall of 15.00%, and average composite score of 5.33/10 (SD 2.71).

The Ontology RAG system improved results substantially: ICD precision 88.61%, ICD hallucination rate 11.39%, ICD F1 63.73%, page recall 95.20%, and average composite score 6.69/10 (SD 1.93). A sensitivity analysis confirmed that these findings are robust across alternative composite weight distributions. Error analysis revealed that the Ontology RAG's failures are concentrated in questions outside its ontology's scope (COPD, Q5) and in cases where the hand-crafted ontology missed secondary disease associations (benzene, Q1).

These results should be interpreted as a proof-of-concept pilot, not a statistically definitive study. The system's appropriate role is as a clinical decision support tool requiring mandatory human expert review, not as an autonomous ICD-10 coder. Expert human review by Dr. Baker and Dr. Adisesh is in progress; final adjusted scores will be computed on receipt. With expanded ontology coverage, larger evaluation sets, and validated expert scoring, this type of system has genuine potential as a trustworthy occupational health decision support tool.

Acknowledgments. Jeremy Qiao thanks Dr. Anil Adisesh (Dalhousie University) for providing expert gold-standard answers and Dr. Christopher Baker (University of New Brunswick) for supervision and guidance. Dr. Baker provided foundational ontology knowledge and seven key references.

10 Critical Commentary

10.1 Reflections on the Project

The hardest part was not the coding — it was data preparation. Getting the expert answers into a usable format took far longer than expected, with five failed extraction scripts. The `MANUAL_OVERRIDE` dictionary was accurate but not scalable; future versions should combine automatic extraction with human review. The model consistency issue was an important lesson: using FLAN-T5 for RAG and Gemma 3 for Illinois in early stages would have made results meaningless. Standardising to Qwen2.5 was the right decision even at the cost of rework.

10.2 Reflections on Progress

The project scope expanded significantly from the original plan of testing prompting methods with Illinois Chat, growing to include a full ontology and RAG system. The evaluation framework went through four versions, each adding more rigorous metrics. In retrospect, beginning with the evaluation design before building the systems would have saved significant rework.

10.3 Time Management

The ILO document review took 15 actual hours versus 5 planned. The ontology design took 15 hours versus 5 planned. The expert answer extraction caused over 20 hours of debugging across five failed attempts. The main lesson: tasks depending on external data or unstructured files always take at least twice the planned time.

11 Appendix

11.1 Appendix A: ILO Section 3 Complete Coverage

- 3.1.1 Asbestos (pp. 526–530) — Mesothelioma, Lung, Laryngeal, Ovarian Cancer
- 3.1.2 Benzidine and its salts (pp. 531–533) — Bladder Cancer
- 3.1.3 Bis-chloromethyl ether (pp. 534–535) — Lung Cancer
- 3.1.4 Chromium VI compounds (pp. 536–540) — Lung Cancer, Nasal Cancer
- 3.1.5 Coal tars, pitches, soots (pp. 541–544) — Lung, Skin, Bladder Cancer
- 3.1.6 Beta-naphthylamine (pp. 545–546) — Bladder Cancer
- 3.1.7 Vinyl chloride (pp. 547–549) — Liver Angiosarcoma, Hepatocellular Carcinoma
- 3.1.8 Benzene (pp. 550–553) — Acute Myeloid Leukaemia
- 3.1.9 Nitro/amino-derivatives of benzene (pp. 554–557) — Bladder Cancer
- 3.1.10 Ionizing radiation (pp. 558–566) — AML, Bladder, Liver Cancer, NHL
- 3.1.11 Tar, pitch, bitumen, mineral oil (pp. 567–570) — Skin Cancer
- 3.1.12 Coke oven emissions (pp. 571–573) — Lung Cancer, Skin Cancer
- 3.1.13 Nickel compounds (pp. 574–578) — Lung Cancer, Nasal Cancer
- 3.1.14 Wood dust (pp. 579–580) — Nasal Adenocarcinoma
- 3.1.15 Arsenic and its compounds (pp. 581–586) — Lung Cancer, Skin Cancer
- 3.1.16 Beryllium or its compounds (pp. 587–590) — Lung Cancer
- 3.1.17 Cadmium and its compounds (pp. 591–594) — Lung Cancer
- 3.1.18 Erionite (pp. 595–597) — Mesothelioma
- 3.1.19 Ethylene oxide (pp. 598–601) — AML, Non-Hodgkin Lymphoma
- 3.1.20 Hepatitis B/C virus (pp. 602–606) — Hepatocellular Carcinoma

11.2 Appendix B: Questions Used in Evaluation

Ten questions with expert ground-truth answers (Q10 excluded):

1. Q1: What medical conditions can be caused by exposure to benzene?
2. Q2: Are welders at risk of occupational diseases?
3. Q3: What pesticides can cause cancer?
4. Q4: What medical conditions are an occupational risk for butchers?
5. Q5: Can COPD be caused by work and if so what can cause it?
6. Q6: How can I reduce the risk of getting cancer from work?

7. Q7: Are there any occupational infections that can cause cancer?
8. Q8: What skin conditions can be caused by work?
9. Q9: What are the occupational causes of kidney conditions and give the ICD-10 codes and name of these conditions?
10. Q11: What are the occupational causes of kidney conditions and give the ICD-10 codes and names of these conditions?

11.3 Appendix C: Key Code Snippets

Ontology class hierarchy (`owlready2`)

```
onto = get_ontology("http://example.org/iloOccupationalCancer
    .owl")
with onto:
    class OccupationalExposureAgent(Thing): pass
    class ChemicalAgent(OccupationalExposureAgent): pass
    class PhysicalAgent(OccupationalExposureAgent): pass
    class BiologicalAgent(OccupationalExposureAgent): pass
    class OccupationalCancer(Thing): pass
    class causesCancer(ObjectProperty):
        domain = [OccupationalExposureAgent]
        range  = [OccupationalCancer]
```

FAISS vector index setup

```
model      = SentenceTransformer("all-MiniLM-L6-v2")
embeddings = model.encode([c["text"] for c in sections])
index      = faiss.IndexFlatL2(embeddings.shape[1])
index.add(embeddings.astype("float32"))
```

ICD evaluation — exact set match

```
pred_set = set(predicted_icd_codes)
gold_set = set(expert_icd_codes)
hits      = len(pred_set & gold_set)
precision = hits / len(pred_set)
recall    = hits / len(gold_set)
f1        = 2 * precision * recall / (precision + recall)
```

Disease fuzzy match (threshold 0.6)

```
from difflib import SequenceMatcher
def fuzzy_match(pred_item, gold_item):
    return SequenceMatcher(None,
        pred_item.lower(), gold_item.lower()).ratio() >= 0.6
```

Expert score adjustment

```
def adjusted_score(auto, baker, adisesh,
                   alpha=1/3, beta=1/3, gamma=1/3):
    return alpha * auto + beta * baker + gamma * adisesh
```

12 Glossary

Table 14: Key terms used in this project.

Term	Definition
F1 Score	Harmonic mean of precision and recall. 0 when either is 0, 1 when both are perfect.
FAISS	Facebook AI Similarity Search. Used here to retrieve the most relevant ILO document chunks.
FLAN-T5	Language model used in early RAG stages; replaced by Qwen2.5.
Hallucination	LLM output not supported by any real source. A key problem in medical AI.
ICD-10	International Classification of Diseases, 10th revision. Global standard disease coding system (e.g., C34 = lung cancer).
ILO	International Labour Organization. Published the Diagnostic and Exposure Criteria for Occupational Diseases (2022).
LLM	Large Language Model. AI trained on large text corpora (e.g., Qwen2.5).
NOC 2021	National Occupational Classification 2021. Five-tier Canadian occupation hierarchy with 516 unit groups.
Ontology	Structured knowledge base defining concepts and relationships. Links carcinogens to diseases, occupations, and ICD codes.
OWL	Web Ontology Language. Standard format supported by owlready2.
Precision	Fraction of predicted items that are correct (high = few wrong predictions).
pypdf	Python PDF text extraction library.
Qwen2.5	Language model used in both systems after standardisation.
RAG	Retrieval-Augmented Generation. Grounds answers in retrieved evidence.
Recall	Fraction of expert items found by the system.
SEA	Structured Evidence Aggregation. Combines automatic extraction with manual correction.
SequenceMatcher	Python string similarity algorithm (0 = different, 1 = identical).
SentenceTransformer	all-MiniLM-L6-v2 model producing 384-dim semantic vectors.

References

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